

LLM-Based Inductive Synthesis of a Local Behavior Tree for Decentralized Multi-Robot Cooperation

Youngsang Kim*, Hyeyun Baek*, Chaeyoung Lim*, Dayeon Hwang*, and Inmo Jang
Korea Aerospace University, S.Korea

* These authors contributed equally to this work



Introduction

Motivation

- Robots sharing the same goal may differ in their positions, roles, and observations; consequently, their control logic is often designed for each robot, which limits scalability as the number of robots or the environment changes.
- Behavior Trees (BTs) provide a modular and reactive control architecture for robot behavior, but designing a separate BT for each robot increases design and maintenance effort and limits reusability.

Objectives

- Achieve decentralized cooperative multi-robot control with a single **Identical BT** that each robot executes using only its local observations.
- Use an LLM to inductively synthesize the *Identical BT* from per-robot demonstrations, improving reusability across diverse tasks and environments.

Methods

Global Plan

- A* computes, for each cube, a path from its start to its goal, which serves as the reference for the Contribution Check.

Identical BT (3 child nodes)

- Contribution Check** – Determines whether the robot's own actions can contribute to the current global plan.
- Assign Check** – When neighboring robots independently select the same task, a decentralized Multi-Robot Task Allocation (MRTA) algorithm resolves the conflict.
- Problem Solving BT** – The LLM-synthesized local action subtree that executes the assigned task.

Two-Stage Inductive Synthesis

- BT Generator:** Guided by the global plan, the LLM generates a subtree for each robot.
- BT Synthesizer:** The robot-specific subtrees are inductively generalized into a single *ProblemSolvingBT* by abstracting task-specific parameters (e.g., target objects, positions, and goals) into blackboard variables, so that every robot can execute the same BT.

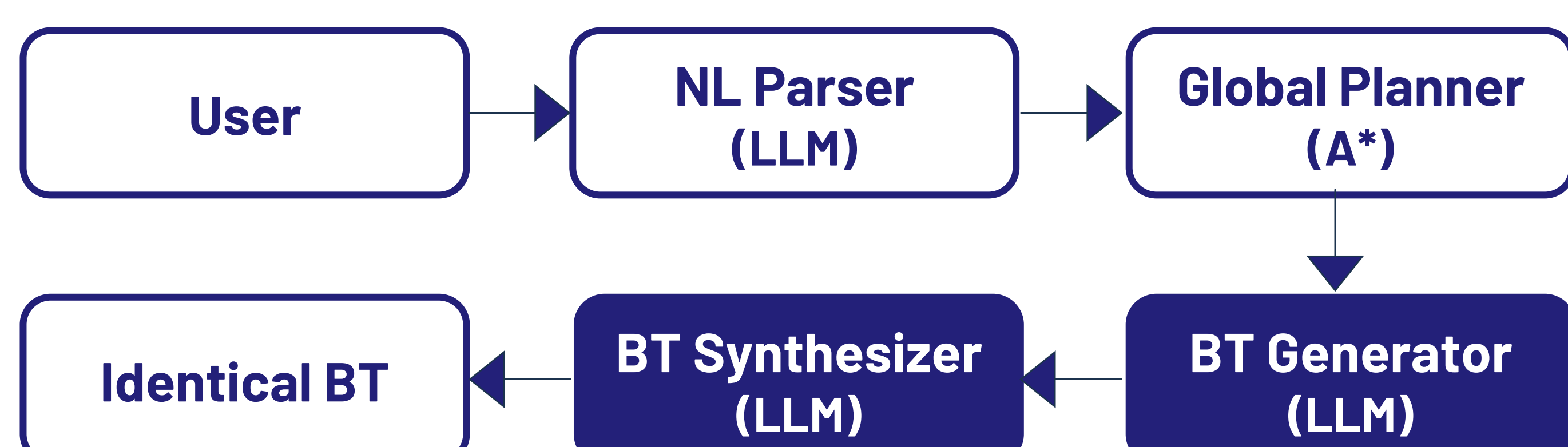


Figure 1. From user command to *Identical BT*

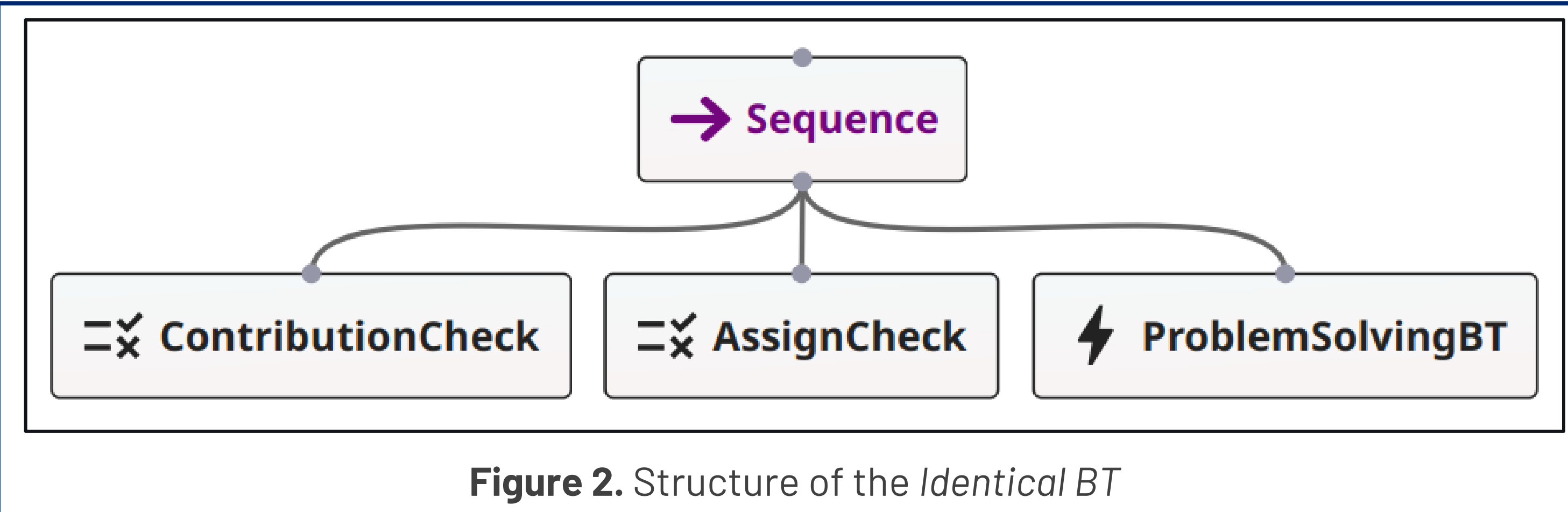


Figure 2. Structure of the *Identical BT*

Experiment

MuJoCo simulator

- ROBOTIS OMX-F robots fixed on an $n \times n$ grid
- Each robot performs pick-and-place only within its adjacent 2×2 cells (Cube Relay scenario).
- Evaluated on 9 conditions = 3 grid sizes (4×4 , 5×5 , 6×6) $\times$ 3 cube counts (2, 3, 4).

One-Step (Baseline)

- A single LLM call directly generates one complete BT (XML) per condition from the same task description.

Results

- A single BT, synthesized from the 4×4 / 2-cube condition in one synthesis call, solved **all 9 conditions (9/9)** without additional LLM calls, whereas One-Step requires a separate generation for each condition and degrades as the number of cubes increases.

Method	2 Cubes	3 Cubes	4 Cubes	Total
One-Step	3/3	2/3	0/3	5/9
Ours	3/3	3/3	3/3	9/9

Table 1. Success by number of cubes (One-Step vs. Ours)

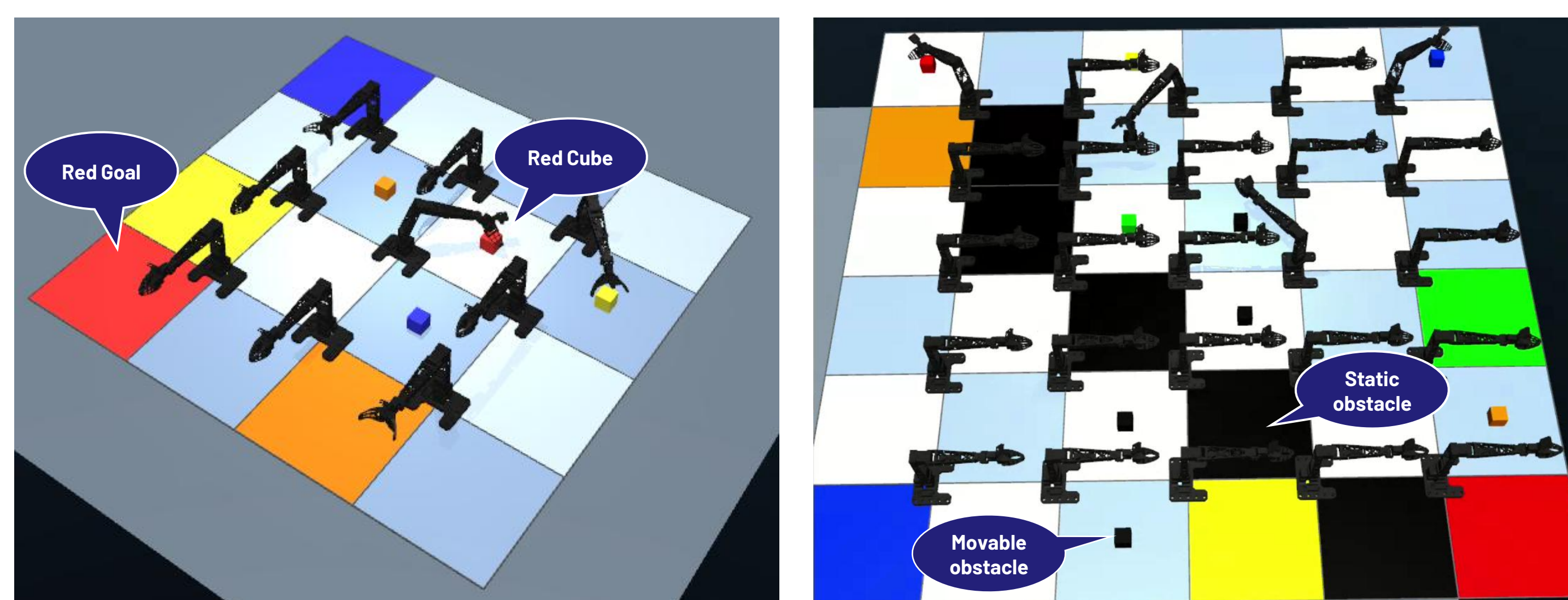


Figure 3. Movable obstacle: goal-less cube that blocks paths but can be relocated.
Static obstacle: blocked cell where no cube can be placed.

Future Work

- Replace A*-based path planning with a domain-independent task planner to generate global plans for arbitrary task domains.
- Evaluate the framework's generalization across a wider range of tasks and environments.
- Deploy the framework on physical OMX-F robots to validate its effectiveness beyond simulation.

